

dea-tools: geospatial analysis of satellite data using Digital Earth Australia, Open Data Cube, and Xarray.

Chad Burton ¹✉

¹ Geoscience Australia, Australia  ✉ Corresponding author

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Summary

The dea-tools package provides: pandas ([pandas development team, 2020](#); ?) and xarray ([Hoyer & Joseph, 2017](#)) Open Data Cube (ODC)'s odc-geo ([odc-geo contributors, 2024](#)).

Statement of need

Satellite remote sensing offers an unparalleled resource for

Features

Analytical tools native to Xarray

Flexible and powerful functionality for scientific analysis of xarray objects: * Extracting time series statistics, using the module dea-tools.temporal: * Generic summary statistics on any time series with temporal_statistics * Phenology with xr_phenology * Linear regression with xr_regression * Validation tools within dea-tools.validation: * Compute common statistical metrics with eval_metric * Generate random points over an xarray object with xr_random_sampling * Spatial analysis tools within dea-tools.spatial: * Vectorise xarray.DataArrays into geopandas.GeoDataFrames with xr_vectorise * Rasterize geopandas.GeoDataFrames into xarray.DataArrays with xr_rasterize * Extract subpixel contours from xarray.DataArrays with subpixel_contours * Interpolate point data stored in a geopandas.GeoDataFrame into an xarray.DataArray with xr_interpolate * Xarray wrappers for the full machine learning pipeline in dea-tools.classification: * prepare data for ML predictions with sklearn_flatten and sklearn_unflatten * Fit models with fit_xr * ML predictions with predict_xr * Handle spatial autocorrelation in cross-validation workflows with spatial_train_test_split and SKCV (spatial k-fold cross validation)

Plotting

The dea-tools.plotting library has extensive functionality for generating beautiful imagery from xarray objects. * RGB plots for true and false colour composite images with rgb * Highly customisable animations with xr_animation

Integration with Open Data Cube libraries

dea-tools works naively with odc-geo

33 **Acknowledgements**

34 Thanks

35 **References**

- 36 Hoyer, S., & Joseph, H. (2017). xarray: N-D labeled arrays and datasets in Python. *Journal*
37 *of Open Research Software*, 5(1). <https://doi.org/10.5334/jors.148>
- 38 odc-geo contributors. (2024). Opendatacube/odc-geo. In *GitHub repository*. GitHub.
39 <https://github.com/opendatacube/odc-geo>
- 40 pandas development team. (2020). *Pandas-dev/pandas: pandas* (latest). Zenodo. [https:](https://doi.org/10.5281/zenodo.3509134)
41 [//doi.org/10.5281/zenodo.3509134](https://doi.org/10.5281/zenodo.3509134)

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